Ashkar A

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.30
SMV Government Model HSS Trivandrum	Class XII (KERALA HSE)	2016	80.00%
SMV Government Model HSS Trivandrum	Class X (KERALA SSLC)	2014	83.00%

SKILLS

Coding: Verilog, SystemVerilog, Python, C, Java

Tools and Environment: Cadence Virtuoso, Cadence Xcelium, Xilinx Vivado, Icarus Verilog, LTSpice, Linux

Domains: RTL Design, FPGA Design, ASIC Design, Timing Analysis, Computer Architecture, AI and Machine Learning

RELEVANT COURSES

Digital Domain: Digital IC Design, CMOS Digital Design, Digital Electronics And Logic Design, Digital IC Design Lab

Analog Domain: Analog IC Design, Analog Electronics, Analog IC Design Lab

Devices and Fabrication: Semiconductor Device Modelling, VLSI Technology

Memory: Memory Device Technology for AI/ML, Memory Circuits and System Design

PROJECTS

M.Tech Thesis | Guide: Dr. Shishir Kumar

AI Accelerator for EMG Classification in Prosthetics | June 2026 - Present

- Developing an AI accelerator for real-time sEMG hand gesture classification for prosthetic control, based on a lightweight and efficient 1D CNN architecture.
- Built a complete software pipeline using the NinaPro DB5 dataset, including EMG preprocessing, window segmentation, normalization, and training/evaluation of a 1D CNN model.
- Designed and analyzed FPGA-oriented hardware architecture for CNN inference, including convolution, activation, pooling, parallelization, and resource estimation for implementation on Zynq-7020.

8-Stage Reconfigurable Multi-Modulus Divider -- RTL Design & Verilog Implementation April - June 2026

- Designed an 8-stage multi-modulus divider using cascaded 2/3 divider cells.
- Extended the divider range to a continuous programmable range of 2-511 using multiplexers.
- Implemented programmable MI masking and frequency selection with RTL

Dynamic Frequency Scaling via Deep Reinforcement Learning Jan - May 2026

- Designed and implemented a DQN-based agent for real-time CPU/GPU clock management on an NVIDIA Jetson embedded platform, targeting frame rate optimization under thermal constraints.
- Engineered a 5-frequency action space with power-bucketed state representation and a shaped reward function balancing FPS target adherence and thermal penalty.

DRAM Array Design & Simulation May 2026

- Designed and simulated a 1T-1C DRAM array, 2-to-4 row decoder, latch-type sense amplifier, and VDD-to-VPP level

shifter using Cadence Virtuoso.
- Implemented and verified write, precharge, read/sensing, restoration, and refresh operations through transient simulation.

WORK EXPERIENCE

IBS Software -- *Senior Solution Engineer*

July 2022 - November 2024 | Trivandrum, India

Product: iFlight - Airline Operations & Crew Management Platform

- Worked as a Software Developer on **iFlight**, a cloud-native airline operations platform used for managing and optimizing flight, crew, and hub operations.
- Developed, enhanced, and maintained software components supporting airline operational workflows and business requirements.
- Participated in the complete software development lifecycle, including requirement analysis, implementation, debugging, testing, and production support.
- Collaborated with cross-functional teams to troubleshoot issues, deliver product enhancements, and maintain the reliability of mission-critical airline software.

POSITIONS OF RESPONSIBILITY

Teaching Assistant (TA) : Embedded Systems

EXTRACURRICULAR ACTIVITIES

Participation in Quiz, Cricket, Football